

Injection scan on the frozen-DINOv2 backbone
(nMSE / free-rollout baseline; warm = worse)

[slot] structpos	1.17× [†] (0.502)	1.02× [†] (0.229)	0.97× [†] (0.463)
[slot] +reweight	0.81× [†] (0.347)	1.05× [†] (0.236)	1.01× [†] (0.485)
[slot] +velocity	0.93× [†] (0.398)	1.01× [†] (0.227)	0.99× [†] (0.473)
[probe] probe	1.02× [†] (0.434)	1.03× [†] (0.231)	1.03× [†] (0.493)
[probe] +slot	0.91× [†] (0.390)	1.10× [†] (0.247)	1.02× [†] (0.487)
[dyn] free MLP	0.99× [†] (0.423)	1.04× [†] (0.234)	1.11× [†] (0.530)
[dyn] strict a=g	1.08× [†] (0.459)	1.16× [†] (0.261)	1.27× [†] (0.610)
[cons] consistency	1.00× [†] (0.428)	1.01× [†] (0.227)	0.96× [†] (0.461)
[free] label-free	0.96× [†] (0.411)	1.00× [†] (0.224)	1.07× [†] (0.512)
[free] grounded	1.11× (0.473)	1.13× (0.254)	1.27× (0.607)
	uniform (both-OOD)	parabola (r/m-OOD)	collision (both-OOD)